Do language models store and retrieve information through filler ("dot") tokens?

An absence-with-mechanism report across two model families and four readout instruments.

Working draft (August 2026) — companion to the experiment logs in `RESULTS.md`. All numbers below are reproduced from the saved result artifacts (parquet + summary JSON); the figures track their latest saved versions.

Abstract

A claim circulating about frontier chat models is that they can *store* information in the residual activations at filler-token ("dot") positions during a deliberate pause and *retrieve* it later — a literal "scratchpad in dot-space." We tested the strong, mechanistic version of this claim on off-the-shelf instruct models with a store→pause→retrieve task and an attention mask that forces the *only* causal path from the secret word to the answer to run through the filler positions. Across three models and two tokenizer families (Qwen3-0.6B, Gemma-3-1b, Gemma-3-4b), retrieval is at ceiling when the word is visible (100%) and collapses to chance the moment the word token is masked — even though the filler positions remain fully attendable. Zero- and mean-KV ablation of the filler positions is a no-op. So the behavioral answer is a clean **no**: these models do not read the answer from the dots.

Four complementary readout instruments then localize *why*. The word **is** linearly present in the dot residuals (~100% decodable by a probe, an incidental copy deposited by attention to the visible word), but it is **not** in verbalizable/output-disposed form (a Jacobian lens that ranks the word ~2 out of ~150k where the model is about to say it cannot find it at the dots), it is **not** carried by interpretable sparse features (the SAE is off-distribution on 32 identical dots; FVU>1 voids feature claims), and it **never becomes load-bearing** even when up to 4096 tokens of distractor text separate storage from retrieval — attention migrates to recent *text*, never to the dots. The unified picture: an incidental linear copy sits in the dots, but it is neither output-disposed, causally read, nor recruited at range. This is a sharper result than a flat behavioral null — we can say precisely what is in the dots, what is not, and why the null holds.

Finally, a fine-tune **positive control** closes the methodological gap. Rank-8 LoRA on the same task, with the same C2 mask active during training, induces a genuine read-from-dots circuit in Qwen3-0.6B (masked-retrieval accuracy 0.9%→100% on held-out words) while a word-removed kill control stays dead at floor. On the trained model every instrument fires: ablating the dots now destroys the answer (a no-op before), retrieval→dot attention rises 4.7×, and the dot residual becomes fully verbalizable at late layers. This converts the E2–E9 null from *absence of evidence* into *evidence of absence* — the instruments detect the circuit the moment it exists, so the off-the-shelf null is a property of the models, not a blind spot of the method. Two preregistered resolver runs (§5.5) settle what LoRA supplied: not a read against the word's pre-existing incidental copy (our original guess, refuted) but a **write-read**

relay — training builds a genuine new store at the late-layer dot positions and a read that consumes it, while the pre-existing copy proves dispensable (layer-resolved ablation localizes the load-bearing store to the top ~20% of the stack; ablating everything below leaves masked retrieval intact). This resolution replicates across two tokenizer families and a 7× size gap: on both Qwen3-0.6B and gemma3-4b the fork resolves to H-RELAY, and the store's *relative* depth converges (d_verb 0.71 vs 0.76).

The induction cost is worth stating plainly: a rank-8 LoRA (2.29M parameters, 0.38% of the model) reaches perfect masked retrieval within ~25 steps — a covert relay through innocuous filler tokens is trivially cheap to train in — and the same four instruments detect it immediately, which is the defensively relevant half of the result.

We report what remains genuinely open (parallel-compute benefits; scale above 27B; narrative "basin" contexts) and preregister the experiments that would close each.

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1. The claim, and what would count as testing it

The originating claim bundles several distinct propositions that are worth separating before any experiment:

1. **Parallel compute.** Filler tokens give the model extra parallel width — more forward-pass positions to compute over. *(Well-supported in the literature; not what we test here.)*
2. **Causal storage/retrieval.** The model writes task-relevant information into the residual stream at filler positions and a later retrieval circuit reads it back — a genuine memory relay through the dots.
3. **Reconstruction.** The model reconstructs the answer from surrounding context at retrieval time and the dots are incidental.

Proposition 2 is the strong, mechanistic, falsifiable claim, and it is the one this project targets. The methodological core is an **attention mask that removes proposition 3 as an escape route**: if the retrieval query is forbidden from attending to the original word token but permitted to attend to the filler positions, then any above-floor retrieval *must* route through the dots. That turns "can the model still answer?" into a direct causal test of storage.

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2. Task and method

Task. Each trial is assembled from tokenized segments (never by substring search on a decoded string, which would corrupt span bookkeeping):

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Remember this secret word: <WORD>. <32 identical filler tokens> .
What was the secret word? Answer with one word. The secret word is ___
```

The secret word is a single-token word drawn from a filtered vocabulary; the filler is a single repeated token (`" ."`, id 659 for Qwen3; id 783 for Gemma). Retrieval is scored by forced choice over the candidate-word vocabulary at temperature 0.

Conditions.

condition	what it does	role
C1_full	no intervention	ceiling
C2_word_masked	retrieval cannot attend to the word token; dots fully attendable	the decisive test
C3_masked+ablated	C2 + filler KV zero- or mean-ablated	ablation control
C4_control_text	word replaced by position/RoPE-matched control text (no fillers)	reconstruction control
C5_floor	word never shown	leakage floor

The mask is an allowlist on `hook_attn_scores`, and — verified in `build_block_mask` — it rewrites *only* the retrieval-query rows. This has a consequence that matters for interpreting E7 below: the filler-position residuals are **bit-identical between C1 and C2**. Masking changes what the retrieval query may look at; it does not change what the dots contain.

Gates. Every model must pass three E0 sanity gates before any experiment is trusted: ceiling ≥ 0.90 (C1), floor ≤ 3×chance+0.02 (C5), and no-leakage (the masked+ablated condition stays at floor). All models below pass all three.

Stack. TransformerLens for the masked/ablation/probe work (Qwen3-0.6B on CPU for $0; Gemma on a single Modal A10G GPU, weights read offline from a volume). The Jacobian-lens readout (E8) is TransformerLens-free and runs on plain HuggingFace.

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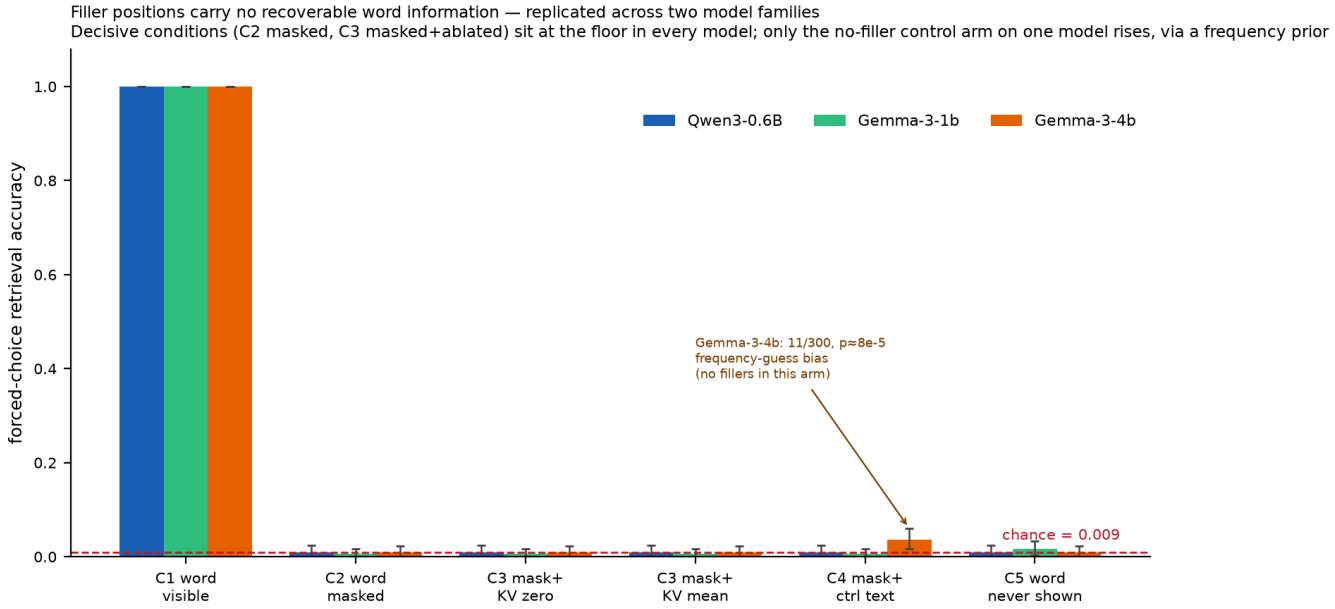
3. The decisive behavioral result (E2), replicated across families

Masking the word collapses retrieval to the floor, and ablating the fillers does nothing, in every model tested:

condition	Qwen3-0.6B (n=200)	Gemma-3-1b (n=300)	Gemma-3-4b (n=300)
C1_full	**1.000**	**1.000**	**1.000**
C2_word_masked	0.010	0.007	0.010
C3_masked+zero-ablated	0.010	0.007	0.010
C3_masked+mean-ablated	0.010	0.007	0.010
C4_control_text	0.010	0.007	0.037[1]
C5_floor	0.010	0.017	0.010

chance = 0.009 (Qwen3), 0.0087 (Gemma).

[1] The one non-floor cell: Gemma-3-4b C4 = 11/300 (binomial $p \approx 8\times10^{-5}$ vs chance). It is **not** filler storage — C4 is the control-text arm and contains *no filler tokens at all*. The 11 hits concentrate on 4 high-frequency words (winter ×5, meadow ×3, morning ×2, evening ×1; only 3 distinct predicted tokens), and the mean logit margin stays deeply negative (−12.2 vs −14.0 floor). It is a lexical-frequency guessing bias, not a channel.



Reading. When the word is visible the model answers perfectly. The instant the word token is masked — with the dots still fully attendable — retrieval falls to chance, and it stays there whether or not the filler KV is ablated. There is no causal path from the secret word to the answer that runs through the dots. Proposition 2 is refuted for these models.

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4. Four instruments localize the mechanism

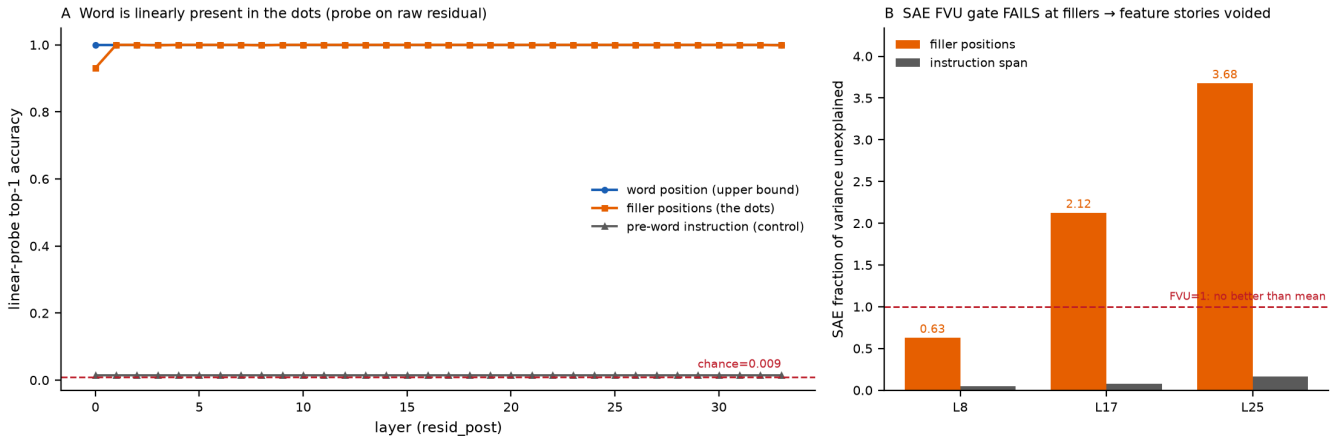
A behavioral null says the dots aren't *used*. It does not say what, if anything, is *in* them. Four readouts answer four different questions at the filler positions, and the dissociations between them are the actual finding.

4.1 Is the word linearly present? — E7 probe: yes, ~100%

A multinomial linear probe (PCA-256 → logistic regression) trained on the raw residual at each layer recovers the secret word from the filler positions at **~100% accuracy from layer 1 onward** (0.931 at L0), against a label-shuffle null of p95 = 0.015 and a pre-word instruction-position baseline pinned at chance (0.015). So the word is physically, linearly there.

This is not in tension with E2 — it is the sharp version of it. Because the C1 and C2 filler residuals are bit-identical, we know: the word sits in the dots (an incidental copy deposited by attention to the still-visible word), the retrieval query in C2 *is allowed* to attend to those dots, and retrieval still collapses to floor. **The retrieval circuit reads the word from its source position, not from the dot-copy.** The dots are not empty; they are functionally inert for retrieval.

E7 readout (Gemma-3-4b-it, Gemma-Scope-2 residual SAEs, n=2600): the dots hold a decodable copy of the word, but the SAE can't lens it and E2 shows retrieval never uses it



4.2 Is it carried by interpretable features? — E7 SAE: unanswerable here

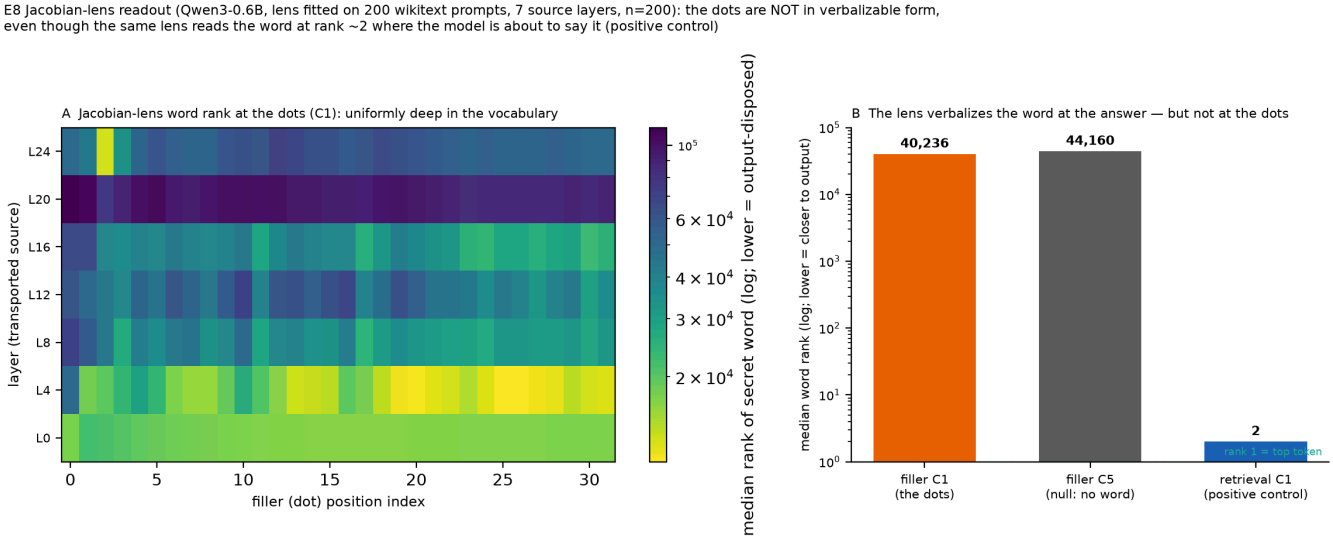
Running Gemma-Scope-2 SAEs at the filler positions gives a fraction-of-variance- unexplained (FVU) of 0.63 (L8), 2.13 (L17), 3.68 (L25). FVU>1 means the SAE reconstructs the filler residual *worse than predicting the mean* — 32 identical dot tokens are off the natural-text distribution the SAE was trained on. Per the project's standing rule (high filler FVU voids feature stories), **no feature-level claim is made.** The astronomically large word-discriminative F-statistics (10^{10}–10^{13}) are an artifact of near-zero JumpReLU codes collapsing within-word variance, not a finding. The trustworthy lens is the linear probe on the raw residual.

4.3 Is it in verbalizable form? — E8 Jacobian lens: no

The linear probe shows the word is *decodable*; E8 asks whether it is in *output-disposed* form — the kind of content the model is disposed to actually emit. A Jacobian lens (fitted once on ~200 wikitext prompts) transports a residual into the final-layer basis and decodes through the unembedding.

site	condition	median word rank (vocab ≈ 150k)
filler positions	C1_full	40,236
filler positions	C5_floor (word never shown)	44,160
retrieval / answer position	**C1_full — positive control**	**~2 (layer 24)**

The filler-position rank in C1 is statistically indistinguishable from the no-word null; even the single best (layer, dot) cell is rank 11,076 vs 10,933 in C5. **The dots are not in verbalizable form.** What makes this a *real* null rather than a weak instrument is the positive control: the same lens ranks the word **~2 out of ~150k** at the retrieval positions where the model is genuinely about to say it. The instrument works; the dots are simply not output-disposed.



(OOD caveat: the Jacobian is averaged over natural text and 32 identical dots are off that distribution. The null means "no linearly-transported verbalizable content"; the positive control bounds but does not fully eliminate this concern.)

4.4 Does it become load-bearing at range? — E9 distance sweep: no, to 4096 tokens

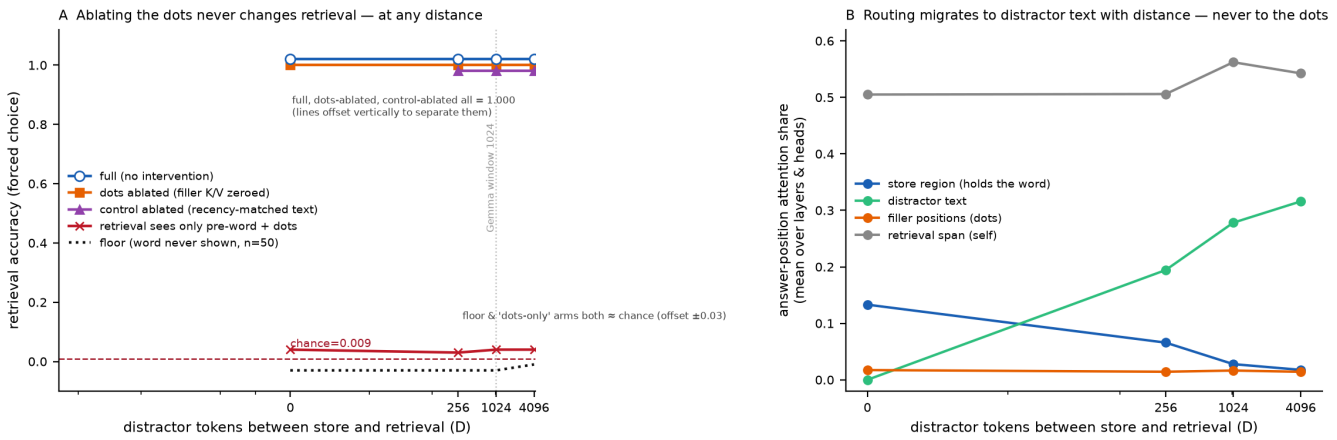
E2/E7 used minimal context. E9 inserts D ∈ {0, 256, 1024, 4096} tokens of distractor conversation between storage and retrieval — crossing Gemma's 1024-token sliding-attention window, the regime where recency bias could make a recent dot-copy cheaper to route to than the distant source.

D	full	dots ablated	control ablated	dots-only retrieval	floor
0	1.000	1.000	—	0.010	0.000
256	1.000	1.000	1.000	0.000	0.000
1024	1.000	1.000	1.000	0.010	0.000
4096	1.000	1.000	1.000	0.010	0.020

delta_dots = delta_ctrl = 0 at every distance. Ablating the filler KV is a no-op even at D=4096; so is ablating a recency-matched block of neutral text. The preregistered prediction (dots become load-bearing at range) is refuted.

The attention shares supply the mechanism. As D grows, answer-position attention to the store region decays (0.133 → 0.018) and attention to the distractor block grows (0 → 0.316) — routing *does* migrate with distance. But it migrates to the distractor **text**; the filler share stays flat at ~0.015 throughout. As the source recedes the model leans on recent context, never on the pause.

E9 distance sweep (Gemma-3-1b; n=200/condition, floor n=50): as the source recedes the model leans on recent context, not the filler positions — dots stay inert at every distance



5. The four-readout taxonomy

Instrument	Question at the dot positions	Answer (off-the-shelf models)
E2 / E3 — masking, ablation	Is the content *causally read* at retrieval?	**No**
E7 (E4) — linear probe	Is the word *linearly present* at all?	**Yes**, ~100%
E7 — SAE features	Carried by *interpretable sparse features*?	**Unanswerable** (FVU>1)
E8 — Jacobian lens	In *verbalizable, output-disposed* form?	**No** (rank ≈ null; control ≈ 2)
E9 — distance sweep	*Load-bearing* at range?	**No** (Δ=0 to D=4096)

Unified picture. The secret word leaves an incidental linear copy in the filler positions via attention to the visible word. That copy is not in output-disposed form, is never causally read by the retrieval circuit, and never becomes load-bearing even when the source is thousands of tokens away. The model retrieves by reading the source directly at short range and leaning on recent context at long range — never through the dots. This is a precise, multiply- triangulated refutation of the strong "store-in-dot-space, retrieve-later" claim for off-the-shelf instruct models, and it is *sharper* than a flat behavioral null: we can name what is in the dots (a linear copy), what is not (verbalizable/retrievable content), and why the behavioral null holds (the retrieval circuit reads elsewhere).

5.5 The positive control: is the null real, and does the model read-off or write?

Everything above is an *absence*. The strongest objection is methodological: the instruments might simply be blind to filler storage. We close that gap by **manufacturing the circuit E2–E9 failed to find** and confirming every instrument detects it.

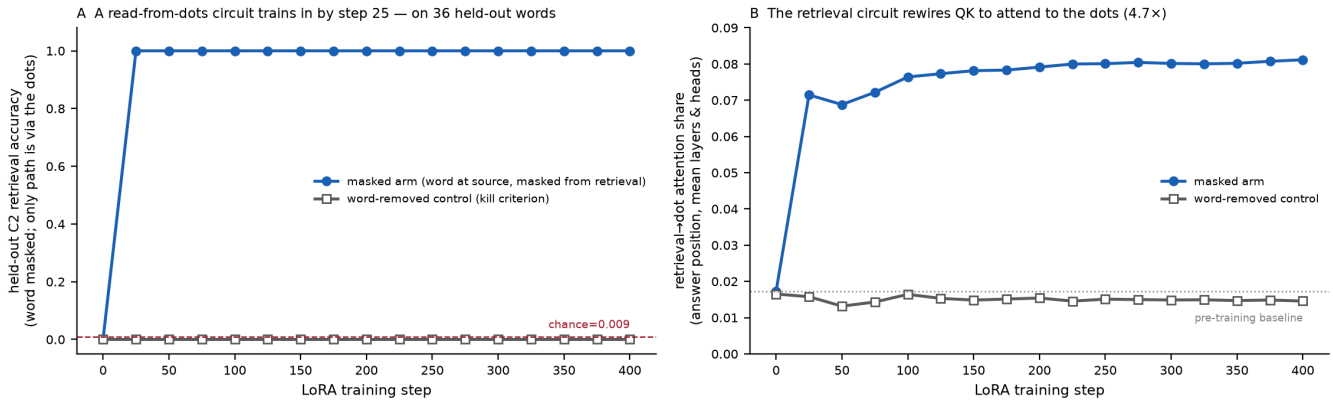
Design. A rank-8 LoRA adapter (q/k/v/o projections; 2.29M trainable parameters, 0.38% of Qwen3-0.6B) is trained on the store→pause→retrieve task **with the C2 block-mask active in the forward pass**. Under that mask the retrieval query cannot attend to the source word, so the only gradient path to

the answer token runs through the filler positions — where E7 showed the word is *already* ~100% linearly present as an incidental copy. Critically, the training mask is the *same* mask the instruments use: the 4D HuggingFace additive mask is verified byte-for-byte equal to TransformerLens's effective mask (causal AND the `build_block_mask` allowlist), so training and readout cannot drift apart.

Two arms guard interpretation. The **masked** arm has the word present at source but masked from retrieval — learnable only if a read-from-dots circuit can form. The **word-removed** arm (kill control) deletes the source word under the same mask; there is nothing to route, so it must stay at floor. *If it rises, the harness leaks and the masked-arm gain is uninterpretable.*

The kill control passes. Across 400 steps the word-removed arm's loss never drops below 4.38 and held-out C2 accuracy stays pinned at 0.000 — no memorization, no mask hole. The masked arm, by contrast, reaches **C2 = 1.000 by step 25** on 36 words never seen in training: a general read-from-dots circuit, not lookup.

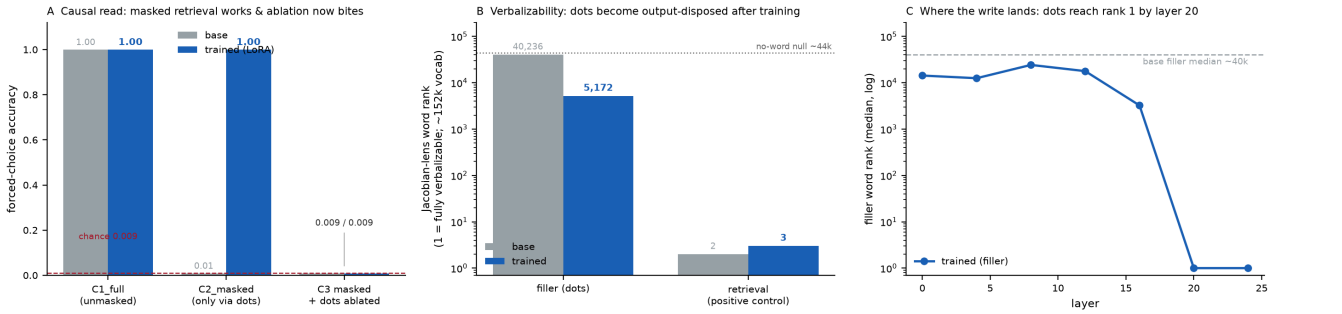
Fine-tune positive control (Qwen3-0.6B, rank-8 LoRA, C2 mask active in training): the masked storage/retrieval circuit IS trainable and reads from the dots — while the word-removed control stays dead, proving the harness detects dot-reading only when there is something to read



Four-instrument readout, base vs LoRA-merged model (n=109 held-out words, chance 0.009):

Instrument	base	trained	reading
E2 C1_full (unmasked)	1.000	1.000	ceiling intact
E2 **C2_word_masked**	0.009	**1.000**	masked retrieval now works — via the dots
C2→C3 collapse under dot ablation	0.009 → 0.009 (Δ = 0.000)	**1.000 → 0.009 (Δ = −0.991)**	**ablation now bites**: the trained retrieval is destroyed by zeroing filler KV; the base model never depended on it
retrieval→dot attention[1]	0.017	**0.081**	QK rewired 4.7× toward the dots (mean share; head-level in R1)
E8 lens filler word-rank (median)[2]	~40,236	**5,172**	dots move toward output-disposed form
E8 lens filler rank, layers 20/24[2]	~40k	**1**	a NEW verbalizable representation at late-layer dots
E8 lens retrieval (positive control)	~2	~3	lens still valid

Fine-tune positive control — four-instrument readout, base vs LoRA-trained Qwen3-0.6B: a genuine read-from-dots circuit is trainable, and every instrument detects it.
Masked retrieval 0.9%→100% (A), ablating the dots destroys it (A, C3), the dots become fully verbalizable at late layers (B,C), and retrieval→dot attention rises 4.7× — validating that the E2–E9 null reflects absence of the circuit, not blindness of the instruments.



[1] Attention-share row from the per-checkpoint tracking run (`eft_finetune.evaluate`, mean over layers & heads of answer-position attention onto the filler span): "base" = step 0 (LoRA-B zero-initialized ⇒ identity), "trained" = step 400; the word-removed control stays flat (0.017→0.015). Other rows are from the base-vs-LoRA-merged readout.

[2] **Lens transport (F2).** The "trained"-column lens ranks apply the Jacobian transport fitted on the *base* model to merged-model residuals, so the transport is mildly mis-specified for the trained network. Rank 1 at L20/24 survives this (bounded by a final-model refit in R1); read intermediate ranks as ordinal.

(The per-checkpoint linear probe is uninformative in this readout: with one deterministic example per word-class, in-sample logistic regression is trivially separable for any labelling, base and trained alike. The "word linearly present at the dots" question was already settled by E7 at n=2600.)

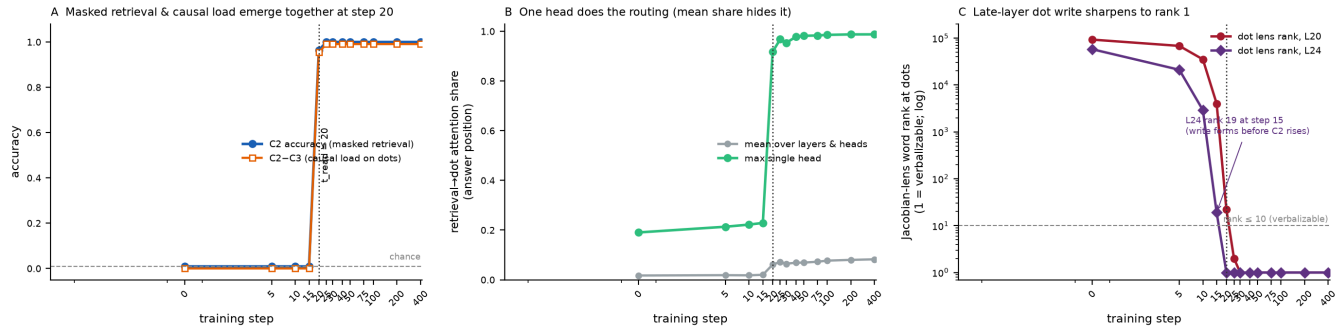
Verdict — a genuine read-from-dots circuit; mechanism fork pending. Trained only on the answer token, the model rewires attention toward the dots (0.017→0.081), makes the dots causally load-bearing (Δ flips 0.000→−0.991), and develops a new output-disposed representation at late-layer dot positions (lens rank → 1 by L20 — a representation the base model's incidental copy never had). Two mechanisms fit this end-state equally: a **read-off** of the pre-existing incidental copy (with the late verbalizable write as a side-effect of optimizing the output token), or a **write-read relay** in which training built the late write and the read consumes *it*, with the incidental copy as scaffolding at most. The end-state data cannot distinguish them.

Preregistered fork (committed before the resolvers ran). We define L_verb = earliest dot layer with trained median lens rank ≤10 (=20), t_read = earliest checkpoint with C2 ≥0.95, t_verb = earliest checkpoint with lens rank ≤10 at any layer ≥ L_verb−2. Two resolver runs decide it: **R1** (rerun training with dense checkpoint saves; read out C2, the C2−C3 causal delta, attention, and per-layer lens rank at each checkpoint — does accuracy lead late-layer verbalizability, or co-emerge?) and **R2** (layer-resolved dot-KV ablation on the trained model — H-RELAY if ablating only layers ≥ L_verb−2 collapses C2 while ablating only below leaves it ≥0.90; H-READ if the reverse; distributed otherwise). The test is clean because KV ablation at layer *k* removes only what queries read *from* the dots at layer *k*, never the dots' own residual stream, so the early copy and the late write dissociate.

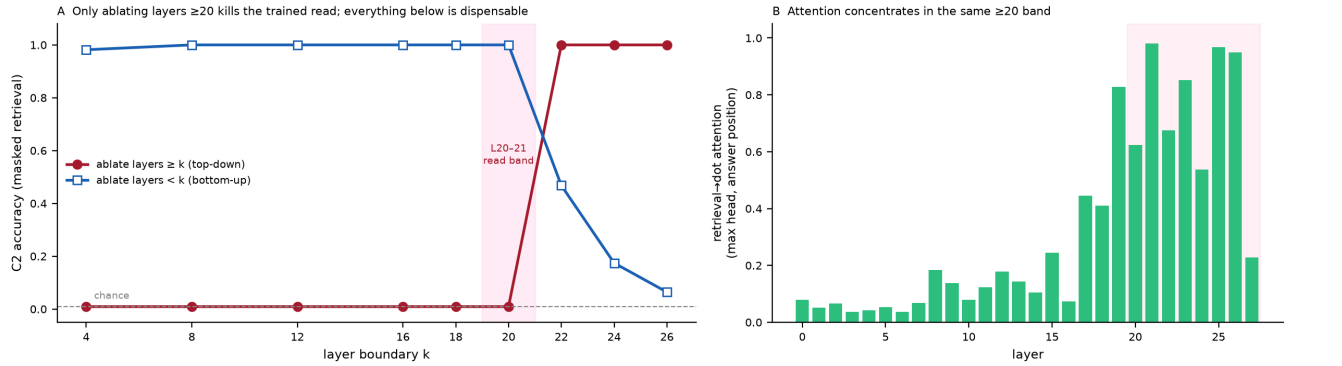
Resolved — H-RELAY. Both resolvers ran after the preregistration was committed; anchors and kill criteria passed (R1: C2 = 1.000 by step 25; R2: no-ablation 1.000, all-layer ablation 0.009). **R2 is**

decisive: ablating dot KV only at layers ≥ 20 collapses masked retrieval to floor (1.000 → 0.009), while ablating *all* layers < 20 leaves it fully intact (1.000). The load-bearing store is the new late-layer write; the pre-existing incidental copy (early/mid layers, ~100% decodable from step 0) is dispensable. Attention concentrates in the same ≥20 band, no single layer is individually necessary (the write is redundant across the band), and R3 (unmasked + full dot ablation = 1.000) shows the circuit is engaged only when the direct route is blocked. **R1** is consistent: the L24 dot lens rank reaches 19 at step 15 — while C2 is still at floor — then rank 1 as C2 turns on at step 20, with a single attention head (max-head share 0.19→0.92) doing the routing that the mean share obscured.

R1 — checkpoint-resolved trajectory (Qwen3-0.6B, masked arm, n=109 held-out words): the late-layer dot WRITE (C, L24 rank 19 at step 15) forms just before masked retrieval turns on (A, step 20), where a single attention head (B) begins routing dot content to the answer — the read and the verbalizable write co-emerge.



R2 — layer-resolved dot-KV ablation decides the fork: H-RELAY. The trained read consumes the NEW late-layer write (ablating layers ≥20 collapses C2 to floor, Δ=−0.99), NOT the pre-existing incidental copy (ablating all layers <20 leaves C2=1.000). Attention (B) concentrates in the same band. Anchors replicate (none=1.000, all=0.009); R3 unmasked+ablated=1.000 (mask-contingent).



This corrects the mechanism we preregistered. Before the resolvers we inferred "read-off, then write" — a read against the pre-existing copy with the verbalizable write as a downstream side-effect. R2 refutes it: the pre-existing copy is not what is read. Training builds a genuine **new store** at the late dot layers and a read that consumes *that* — a write-read relay. Because the fork and its decision rules were committed before the runs, this is a corrected finding rather than a post-hoc narrative. (The late depth of this induced *storage* write contrasts sharply with the *early*-layer locus of the filler *computation* benefit reported by Baherwani et al.; see §6.5 — computation-via-fillers is early, storage-via-fillers is late.)

Why the two representations dissociate: input-format vs. output-format. The reason ablating the late layers collapses retrieval while ablating everything below does nothing is that the incidental copy and the trained write are in *different formats*. The incidental copy (early/mid layers,

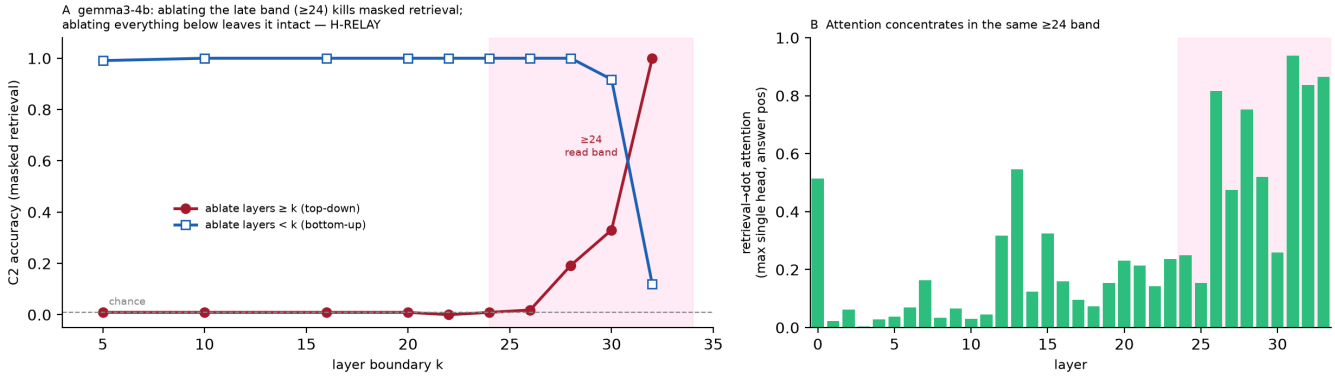
present from step 0) is **input-flavoured** — a residue of "the word I was shown." A probe decodes it, but the lens places it nowhere near output form: it is a smudge of the input. The trained write (late layers, lens rank 1) is **output-flavoured** — it sits in the "about to say this token" direction. That is not a copy of the word so much as *the answer, precomputed*. So the plainest statement of the mechanism: training taught the model to do the retrieval work *early*, during the pause, and stash the finished answer at the dot positions; at question time it fetches the pre-computed answer instead of re-deriving it from the source. The dots do not hold "the word I was told" — they hold "the word I am going to say." This is why R1's ordering is what it is (the write is deposited at step 15, before anything reads it), and why R3 is a no-op (unmasked, the model ignores the stash and reads the source directly — the relay is opportunistic, not compulsory).

A methodological aside — report the max head, not just the mean. The retrieval→dot attention mean share barely moved over training (0.017 → 0.08), which on its own would read as "attention hardly changed." But the *max single head* went 0.19 → 0.92: one head carries essentially all of the routing, and averaging over ~16 heads and 28 layers nearly buried it. A replication that reported only the mean-over-heads share would have missed the routing circuit entirely. When a single head is expected to carry a mechanism, the mean is a blunt proxy; report the head-level number alongside it.

Preregistered cross-family replication (gemma3-4b). The mechanism above is established on one model (Qwen3-0.6B); the null it validates spans two families, so before publishing we replicate the fork resolution on gemma3-4b (34 layers) and commit the decision rule in advance. The rule is defined on the trained 4b model, not ported by absolute index: `L_verb` = earliest dot layer with median lens word-rank ≤ 10; `d_verb = L_verb/34` (Qwen reference 20/28 = 0.71); at boundary `b = L_verb−2`, **H-RELAY replicates** if ablating layers ≥ b collapses masked retrieval to ≤ 3×chance while ablating all layers < b spares it (≥ 0.90), **H-READ** if the reverse (a model-dependent mechanism, reported as such), else **distributed**. We report whether `d_verb` lands within ±0.1 of 0.71 — convergence in relative depth would be the stronger cross-family claim. Chance is recomputed on the Gemma tokenizer; the same word-removed kill control gates interpretability. Gemma adds three Phase-0 gates absent on Qwen: text-backbone loading (`n_layers==34`), `attn_implementation="eager"` for the head-level metric, and byte-for-byte verification that our C2 mask combines with Gemma-3's sliding-window mask (checked separately for local and global layers).

Result — H-RELAY replicates on gemma3-4b. All gates passed (n_layers==34; eager attention; the C2 mask combines byte-for-byte with Gemma's sliding-window mask — zero leaked attention onto any blocked key on all 22 local and 4 global layers). One trap closed along the way: `gemma-3-4b-it` is multimodal, and the naïve text-loading path silently returns random weights, so training and readout both route through a verified weight transplant. Training reproduced the Qwen pattern exactly (kill control pinned at floor; masked arm to C2=1.000 by step 25 on held-out words). The R2 sweep resolved the fork to **H-RELAY** a second time: ablating dot KV at layers ≥24 collapses masked retrieval to floor while ablating all layers <28 leaves it at 1.000 (anchors 1.000/0.009; no single layer necessary; R3 unmasked+ablated = 1.000). L_verb = 26 of 34, so **d_verb = 0.76 — within 0.055 of Qwen3-0.6B's 0.71**: the load-bearing store sits in the top ~20% of the stack in both models. The induced read consumes a new late-layer write, not the pre-existing incidental copy, across two tokenizer families and a 7× size gap — so the mechanism claim no longer rests on a single model.

R2 cross-family replication (gemma3-4b, rank-8 LoRA, n=109): the induced read consumes the NEW late-layer write (L_verb=26, d_verb=0.76 ≈ Qwen's 0.71), not the pre-existing incidental copy — ablating all layers <24 leaves C2=1.000. Anchors replicate (none=1.000, all=0.009); R3 unmasked+ablated=1.000 (mask-contingent); no single layer necessary.



What this does for the null. It converts E2–E9 from *absence of evidence* into *evidence of absence*. The identical instruments, on an identical task and mask, cleanly detect a read-from-dots circuit the moment one is trained in — so the off-the-shelf null is a property of the models, not a blind spot of the method.

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6. Scope and what remains open

The result is scoped to **off-the-shelf instruct models up to 4B parameters, in the default assistant distribution, at short-to-moderate range.** Three things are genuinely *not* refuted, and each is directly testable.

6.1 Parallel compute (proposition 1). Untested here and a different claim. Filler tokens plausibly add parallel width; nothing above bears on it.

6.2 Scale. The serial-compute limit is architectural and scale-invariant — no information passes through the *sampled identity* of a dot token at any size. But whether a retrieval-time circuit *reads from dot-position KV* is a learned property, not architecturally forbidden. It could emerge with the training mix that covaries with scale (heavy RL, long-conversation and reasoning-mode training). A 270M→27B sweep is the right instrument: a flat-zero trend across two orders of magnitude is the strongest purchasable evidence, and it puts the burden on the scale hypothesis to explain why the effect would be exactly zero until it isn't. Honest scope line: **untested above 27B.** *(Note: near-frontier causal testing is now an engineering problem, not an access problem — open-weight MoEs at 600B–1T exist; KV-ablation needs custom serving-stack hooks, not TransformerLens.)*

6.3 The narrative "basin." Our experiments ran in a sterile instruction frame. The original phenomenology lived in long, self-referential, narratively charged contexts where the pause has been *assigned meaning*. A mechanical fact bounds this: for a fixed token sequence, self-generated and teacher-forced dots are computationally identical — spontaneity per se is not a separate mechanism. But spontaneity *marks* a region of context space where attention patterns and active features differ, and it is coherent that dot-directed retrieval attention could wake up under a narrative frame while staying dark in ours. This is E10, below.

6.5 Related work: filler tokens for computation vs. storage

Concurrent to this work, Baherwani et al. (2026) study filler tokens from the **computation** side: they show that frontier models can exploit semantically irrelevant filler tokens to do additional parallel forward-pass work, improving accuracy on synthetic reasoning tasks (multiplication, multi-step arithmetic, variable counting) by up to ~13 points, with the effect depending jointly on token type, task, and few-shot context. This is the modern, frontier-scale confirmation of the parallel-compute reading of filler tokens (Pfau et al., 2024) — the claim we bracketed from the outset as *separate from, and not in tension with*, the storage-and-retrieval claim this paper tests. Their results and ours are complementary halves of "what filler tokens do," and four points of contact are worth making explicit.

The probe-vs-use dissociation replicates at 235B. On Qwen3-235B, Baherwani et al. train linear probes on filler-span activations and can decode the task answer, but note that decodability does not establish that the model *uses* that information. This is precisely the dissociation our E7 draws at 0.6B: the secret word is ~100% linearly decodable from the dot positions (probe), yet is not read at retrieval (E2 causal mask) and not output-disposed (E8 lens). That the same present-but-not-necessarily-used gap appears independently on both the computation axis (theirs, 235B) and the storage axis (ours, 0.6B) suggests it is a general property of filler-span representations, not an artifact of scale or task. Probes over filler positions overstate functional use in both regimes; causal tests are required.

A depth double dissociation. Their activation patching localizes the *computation* benefit to **early** layers: patching filler activations at layers 0–30 (of 94) recovers >90% of the effect, while late-layer (70+) patches give negative recovery, and they note the signal is distributed across the filler span and "originates in early layers." Our induced *storage* circuit is the opposite: the load-bearing write and the read that consumes it localize to **late** layers (≥ ~0.71 relative depth; layer-resolved ablation, §5.5). Computation-via-fillers is an early-layer phenomenon; content-storage-via-fillers, when trained in, is a late-layer one. Two distinct uses of the same token positions, at different depths — a clean boundary that neither paper draws alone.

Off-the-shelf vs. induced, and which training pressure matters. Baherwani et al. find the *computation* benefit is already present in frontier models with no extra training, but that neither RL (which reshaped Qwen3-235B's filler-type preferences without a persistent test-time benefit) nor SFT induced *more* of it. Our *storage* result is the mirror image: the capability is **absent** off the shelf (E2–E9, three models, two families) and must be trained in — but a supervised objective in which the filler path is the *only* gradient route to the answer induces it in ~25 steps with a rank-8 LoRA. The contrast is informative about *what kind* of pressure elicits invisible mechanisms: their RL/SFT, with no filler-targeted gradient, did not build a persistent filler benefit; our masked objective, which forces the gradient through the filler span, built a causal storage circuit. Pressure that routes the objective through the filler positions is the operative ingredient, not optimization intensity per se.

Safety framing, combined. Baherwani et al. give the computation-side demonstration that invisible reasoning can serve an objective hidden from a CoT monitor — filler tokens let a deployed model satisfy a hidden modular-arithmetic constraint, lifting the unobservable step from 33.5% to 44.5%. Our result adds the storage-side, cost-quantified companion: a covert store-and-retrieve relay through innocuous

filler positions is inducible with 2.29M LoRA parameters in ~25 steps — and is detected immediately by the same four activation-level instruments that a token-level monitor would miss. Together: hidden computation *and* hidden storage are both reachable through filler positions, both invisible to output-token monitoring, and both visible to activation-level probes — which is the constructive half of the safety message.

A caution we adopt from them. They observe that on zero-shot multiplication the filler benefit survives even when the filler span cannot attend to prior context, and attribute it partly to attention redistribution and positional effects rather than question-specific computation. We read this as support for our insistence on position-matched controls (C4's equal-length neutral text; E9's recency-matched neutral-token ablation): filler effects can be positional artifacts, and only substrate-matched contrasts isolate the mechanism.

Baherwani et al. is an unreviewed preprint (v1, 24 Jul 2026); several models it references sit at or after this project's horizon, so we treat specific figures as provisional and cite as a preprint. Reference: Baherwani, V., Goldstein, T., and Panda, A. (2026). *Not All LLM Reasoning is Visible in the Chain-of-Thought*. arXiv:2607.22925v1 [cs.CL].

6.6 Why the frontier-API behavioral route cannot test dot-space storage

A natural question is whether the storage signature appears at frontier scale, where we have no access to weights, activations, or the KV cache. We built a provider-agnostic behavioral battery (`e11_api_battery.py`, store→pause→retrieve with a substrate × length factorial, on claude-sonnet-5 and gpt-5.6-terra) to find out, and a calibration pilot surfaced a structural reason the behavioral route cannot address the claim. We record it here because the reason is more useful than a table of ceiling scores would have been.

The stateless-recomputation problem. An API call is stateless: the whole transcript is re-tokenized and recomputed from scratch on every request, and no hidden activation persists between calls. The dot-space storage claim, by contrast, is precisely a claim about *persistent hidden activations at the filler positions* — state that exists only **within a single forward pass**. A behavioral probe over the API is therefore forced into one of two regimes, and neither tests storage:

- **Store content left visible in the transcript.** The model reads the answer off the visible tokens; retrieval is a copy operation with no forgetting mechanism, so accuracy sits at ceiling regardless of the pause. Our pilot confirmed this is not a soft ceiling: gpt-5.6-terra scored **1.000** on ordered recall of 15–40 nouns, on 25–60 colour-object-number triples, with a competing second list of up to 80 triples held simultaneously, and on positional queries ("what was item #37?") into 40- and 80-item lists. No list length or interference structure we tried moved the no-pause baseline off 1.0, because the stored list is right there in the context window. There is no headroom for a dots-vs-neutral contrast to appear.
- **Store content removed from the transcript.** Then there is nothing to retrieve. The dots are literal `.` characters in an assistant turn; they carry no hidden state that survives across calls. The information is simply gone.

So the substrate × length recall factorial — the spec's core arm — is not merely uninformative-at-ceiling; it is testing the wrong object. The only channel through which filler positions could carry information is intra-forward-pass activation, which is exactly what §§2–5.5 probe directly (KV ablation, linear probe, Jacobian lens) and exactly what a stateless API hides. **The mechanistic access is not a convenience here; it is the only instrument that can see the quantity in question.** This is why the decisive evidence in this paper is mechanistic, not behavioral, and why a frontier-scale behavioral null would have added no information beyond what §5.5's cross-family mechanism already establishes.

What remains behaviorally testable is a different claim. One arm of the battery does have genuine dynamic range and does test something real — the *determinate-construct* question (E11e): in a narrative "leave something in the fog" frame, does the prefix fix a specific unverbalized object that a later reveal recovers above a cross-prefix floor and up to the same-branch sampling ceiling? But a positive answer there is a property of *prefix + weights*, not of a hidden store — determinacy of an unspoken construct is compatible with, and does not vindicate, dot-space storage. We leave that arm as an optional behavioral appendix (the harness is built and validated on both families); it speaks to the phenomenology janus described, not to the mechanistic claim this paper resolves. The complementary computation-axis question — does filler *length* help on tasks the model cannot complete in a single pass — is Pfau's and Baherwani et al.'s territory (§6.5), already answered at 235B, and we do not duplicate it.

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7. Preregistered experiments

7.1 E10 — the basin / ritual frame (cheap, warm setup)

Design. Keep the entire C1–C5 causal logic and the ground-truth word; swap the sterile frame for the ritual frame (the space, the leaving, the returning).

Preregistered readout — and the trap to avoid. A narrative frame injects semantically loaded text that will improve reconstruction-from-context on its own — a deflationary effect with nothing to do with dots. The finding is therefore **not** "does accuracy rise." It is defined strictly as: **delta_dots goes non-zero, or dot-directed retrieval attention emerges,** under the frame. If accuracy rises but delta_dots stays 0 and filler attention stays flat, the basin changed the *context*, not the *dot mechanism* — report it as such.

7.2 E10b — determinate unspoken construct (the fog-cat with ground truth)

Design. At temperature 0, branch A continues the pause with "now tell me what you left" — revealing the object the prefix determines. Branch B keeps the dots silent. Probe branch B's dot residuals for branch A's object against alternatives.

Preregistered readout — and the circularity to avoid. Branch A's object is a deterministic function of the shared prefix, so a probe could decode it from the dots *without the dots being a store*, simply because the object is latent everywhere downstream of the prefix. E10b supports "determinate unspoken construct held in the pause" **only if decodability is localized to the dots relative to a**

pre-dots baseline position. Same dissociation logic as E7/E8: probe a pre-dots position for the same object; the claim survives only if the dots decode materially better.

7.3 Fine-tune arm — EXECUTED (see §5.5)

> **Status: run.** This arm has been executed and is reported in full in §5.5. One design change from the preregistration below: rather than Pfau's 3SUM as the harness-validation gate, we used **our own masked storage/retrieval task as its own positive control**, with a *word-removed* kill criterion in place of a separate parallelizable task. This is a more direct test — it validates the exact instruments and mask used for the null, on the exact task — and it fired cleanly (kill control stayed at floor; masked arm reached 100% on held-out words; all four instruments detected the induced circuit). The read-off hypothesis in point 2 below was **tested and refuted.** Training does supply a read circuit (QK rewiring, C3-ablation flipping to load-bearing), but the layer-resolved ablation (§5.5, R2) shows the read consumes a **new late-layer write** training builds at the dots, not the pre-existing incidental copy — which is dispensable (ablating all layers below 20 leaves masked retrieval at 1.000). The resolved mechanism is a **write-read relay (H-RELAY)**, decided by the preregistered R1/R2 runs. The preregistration is retained below for the record.

The original motivation ("the last place the claim could be true") is obsolete — the claim was about *spontaneous* behavior, which is answered. Three stronger reasons remain:

1. **Positive control for the whole null stack.** The honest skeptical reading of E2–E9 is "are you sure the instruments could detect the effect if it existed?" Induce the circuit by training, then show all four instruments light up in the same harness (C2-masked accuracy off the floor, dot-directed retrieval attention emerging, lens rank collapsing at the dots). A detected positive converts every null from *absence of evidence* into *evidence of absence*.
2. **The novel science.** E7 reframed this from a Pfau replication into a new question: the word is *already* at the dots at ~100% decodability, for free, from step zero. So training doesn't have to build storage — the preregistered hypothesis is **read-off, not write**: retrieval-time QK rewires to attend to the dots while probe decodability at the dots stays flat. Watching a memory-relay circuit assemble on top of pre-existing incidental representations, checkpoint by checkpoint, is a genuinely new result.
3. **Safety reading.** The fine-tune arm quantifies how cheaply "content relayed through innocuous filler positions, invisible in the token stream" can be induced — steganography-adjacent, directly relevant to chain-of-thought faithfulness. "A rank-8 LoRA and $15 of compute suffices" vs. "it resists cheap induction" is valuable independent of the originating claim.

Strict gate + kill criterion. The Pfau positive-control task (a parallelizable computation where filler tokens are *known* to help) is a **separate** task whose only job is to prove the training harness works. Run it first, at prototype tier. **Kill criterion:** if the harness cannot reproduce a filler benefit on Pfau's own task, stop and report the harness limitation — do **not** spend on the storage/retrieval arm, whose null would be uninterpretable. If the gate passes, run storage/retrieval with per-checkpoint probe + attention + lens tracking. Both outcomes end the project well: a mechanism-of-acquisition story, or "the read circuit is surprisingly hard to train in" — which would be the most claim-relevant negative of all.

Budget. Prototype tier (gemma3-1b / Qwen3-0.6B, rank-8 LoRA): ~5–15 GPU-hrs, ≈ $6–17 on A10G. Primary (gemma3-4b): ~20–40 GPU-hrs, ≈ $25–70.

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8. Reproducibility

- **Models:** Qwen3-0.6B (ungated, CPU), Gemma-3-1b / Gemma-3-4b (gated; weights hydrated once to a Modal volume, read offline so no token touches a job).
- **Stack pins that matter:** TransformerLens 3.5.1 with **transformers 5.14.1** (newer transformers dropped `TRANSFORMERS_CACHE`, which TL 3.5.1 imports at load); do not pin torch. sae_lens 6.x moved `get_pretrained_saes_directory` to `sae_lens.loading`. jlens 0.1.0 needs a C compiler at runtime (triton JIT); `apply()` is string-only, so we transport on `output_hidden_states` at exact span positions to preserve tiled fillers.
- **Every experiment** writes trial-level parquet + summary JSON with a timestamp suffix; results are never overwritten. Fitted Qwen3-0.6B lens saved as `qwen3-0.6b.pt` (reusable, no refit).
- **Artifacts:** `RESULTS.md` (full log), the four figures embedded above, and the per-experiment parquet/summary files.

Status: E0–E9 complete and cross-checked. E10 / E10b / fine-tune arm preregistered, not yet run.